

### Calc 3 – Vectors & Geometry of $\mathbb{R}^3$

#### *Worksheet A*

1. For  $\mathbf{u} = \langle 1, 2, -1 \rangle$  and  $\mathbf{v} = \langle 3, 0, 4 \rangle$ , compute  $\mathbf{u} \cdot \mathbf{v}$  and the angle between them.
2. Compute  $\mathbf{u} \times \mathbf{v}$  for the same  $\mathbf{u}, \mathbf{v}$  as above.
3. Find the equation of the plane through  $(1, 0, 2)$  with normal  $\langle 2, -1, 3 \rangle$ .
4. Find parametric equations for the line through  $(0, 1, 2)$  and  $(3, 4, -1)$ .
5. Find the distance from the point  $(2, 1, 3)$  to the plane  $x + 2y - z = 5$ .

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#### *Worksheet B*

1. For  $\mathbf{a} = \langle 2, -1, 3 \rangle$  and  $\mathbf{b} = \langle 1, 4, 0 \rangle$ , find  $\text{proj}_{\mathbf{b}} \mathbf{a}$ .
2. Find the area of the triangle with vertices  $A = (1, 0, 0)$ ,  $B = (0, 2, 0)$ ,  $C = (0, 0, 3)$ .
3. Find the equation of the plane through  $A, B, C$  from problem 2.
4. Find the angle between the planes  $x + y + z = 1$  and  $2x - y + z = 3$ .
5. Identify the quadric surface  $4x^2 + 9y^2 - z^2 = 36$ .

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Answer Key (Worksheets A & B)

#### Worksheet A.

1.  $\mathbf{u} \cdot \mathbf{v} = 3 + 0 - 4 = -1$ .  $\|\mathbf{u}\| = \sqrt{6}$ ,  $\|\mathbf{v}\| = 5$ .  $\cos \theta = -1/(5\sqrt{6})$ , so  $\theta = \arccos(-1/(5\sqrt{6})) \approx \boxed{94.7^\circ}$ .
2.  $\mathbf{u} \times \mathbf{v} = \det(\mathbf{i}(2 \cdot 4 - (-1) \cdot 0) - \mathbf{j}(1 \cdot 4 - (-1) \cdot 3) + \mathbf{k}(1 \cdot 0 - 2 \cdot 3)) = \boxed{\langle 8, -7, -6 \rangle}$ .
3.  $\boxed{2(x-1) - (y-0) + 3(z-2) = 0}$  i.e.  $2x - y + 3z = 8$ .
4. Direction  $\langle 3, 3, -3 \rangle$  (or  $\langle 1, 1, -1 \rangle$ ).  $\boxed{x = 3t, y = 1 + 3t, z = 2 - 3t}$ .
5.  $D = \frac{|2 + 2 - 3 - 5|}{\sqrt{1 + 4 + 1}} = \frac{4}{\sqrt{6}} = \boxed{\frac{4\sqrt{6}}{6} = \frac{2\sqrt{6}}{3}}$ .

#### Worksheet B.

1.  $\text{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{b} \cdot \mathbf{b}} \mathbf{b} = \frac{2 - 4 + 0}{1 + 16 + 0} \langle 1, 4, 0 \rangle = \boxed{\langle -2/17, -8/17, 0 \rangle}$ .
2.  $\mathbf{AB} = \langle -1, 2, 0 \rangle$ ,  $\mathbf{AC} = \langle -1, 0, 3 \rangle$ .  $\mathbf{AB} \times \mathbf{AC} = \langle 6, 3, 2 \rangle$ . Area =  $\frac{1}{2} \sqrt{36 + 9 + 4} = \boxed{\frac{7}{2}}$ .
3. Normal =  $\langle 6, 3, 2 \rangle$ , through  $A = (1, 0, 0)$ :  $\boxed{6(x-1) + 3y + 2z = 0}$  i.e.  $6x + 3y + 2z = 6$ .
4. Normals  $\mathbf{n}_1 = \langle 1, 1, 1 \rangle$ ,  $\mathbf{n}_2 = \langle 2, -1, 1 \rangle$ .  $\cos \theta = \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|} = \frac{2}{\sqrt{3} \cdot \sqrt{6}} = \frac{2}{3\sqrt{2}} = \frac{\sqrt{2}}{3}$ .  $\theta = \arccos(\sqrt{2}/3) \approx \boxed{61.9^\circ}$ .
5. Divide by 36:  $\frac{x^2}{9} + \frac{y^2}{4} - \frac{z^2}{36} = 1$ . **Hyperboloid of one sheet** (axis along  $z$ ).